

Pavitra Jain

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I am an undergraduate student in Computer Science (AI & ML) at PES University with a strong foundation in applied mathematics and a genuine interest in ML theory and research. I enjoy working on problems that sit at the intersection of rigorous mathematics and real-world systems, and I am actively seeking research opportunities in applied AI and computational sciences.

Education

PES University	Bangalore, India
B.Tech in Computer Science (AI & ML)	2024 – 2028
CGPA: 8.85	
Class XII: 93% — Class X: 96.2%	

Research Interests

Machine Learning Theory
Optimization Algorithms
Applied Linear Algebra in Neural Networks
Number Theory
String Theory

Hackathon Awards

Tech Solstice 2026, MIT Bengaluru	March 2026
Prize: Rs. 1,00,000 36-hour hackathon, 80+ competing teams	
Built a real-time AI observability system for financial fraud detection featuring a six-model ensemble (graph-based risk, sequence anomaly detection, behavioral biometrics, tabular ensemble, AML scoring, and language-based fraud analysis), containerized with Docker and monitored via Prometheus, Loki, and Jaeger.	

Vyuhatech 2.0 National Level Hackathon, CMRIT	2025
400+ online participants, 75 software teams in final round	
Built VISIONX, a missile command logging system combining Random Forest anomaly detection, real-time WebSocket supervisor alerts, and blockchain-based immutable audit logging to prevent aerospace mission failures caused by human input errors. Stack: Python, TensorFlow/PyTorch, React.js, Hyperledger/Ethereum.	

Mystara 2025 Hackathon, Cyber Security Track	October 2025
24-hour competition evaluated by a panel of seven industry judges	
Developed a full cybersecurity solution from scratch within 24 hours as part of a six-member team, covering threat detection and secure system design under competitive constraints.	

Sahay AI Vichar Ideathon	March 2026
Designed a renewable energy verification system integrating IoT-based smart meter monitoring,	

AI-driven solar output prediction, anomaly detection for fraudulent reporting patterns, and a gamified incentive layer to promote accountability in clean energy systems.

Intentia '26, PES University Psychology Hackathon

March 2026

Built *SentinelMind*, an early mental health detection system using an Isolation Forest model trained on 100,000 student profiles across 28 behavioral signals (attendance, LMS activity, sleep, social withdrawal). Integrated PHQ-9/GAD-7 clinical screening with a mentor dashboard and a self-recalibrating feedback loop.

Projects

SentinelMind – Student Mental Health Early Warning System

Developed an AI-powered behavioral monitoring system using Isolation Forest trained on 100,000 student profiles across 28 signals. Integrated clinical screening (PHQ-9/GAD-7), a mentor dashboard, and a self-recalibrating feedback loop. Built with Python, scikit-learn, Flask, VADER NLP, and Vanilla JS.

Renewable Energy Verification System

Designed an end-to-end clean energy accountability system combining IoT-based smart meter monitoring, AI-driven solar output prediction, anomaly detection for fraudulent patterns, and a gamified incentive layer for transparent energy reporting.

Satellite Command Validation using Anomaly Detection

Designed and implemented an ML-based anomaly detection system for satellite telemetry validation. Applied supervised learning techniques and statistical thresholding to identify irregular command behavior.

Deep Learning for X-Ray Pathology Detection

Developed a CNN-based diagnostic model capable of classifying 18 medical conditions from chest X-ray datasets. Focused on preprocessing, feature extraction, and model generalization.

Technical Skills

Programming: Python, C

Machine Learning: TensorFlow, PyTorch, scikit-learn

Mathematics: Differential Calculus, Linear Algebra, Multivariable Calculus, Permutations and Combinations

Honors

C.N.R. Rao Scholar (3x) – Awarded for academic excellence across consecutive semesters

Academic Activities

Core Member - Shunya Mathematics Club

PR & Media Team – ARCH Club

Hospitality Organizer—Samarpana India